APM1110 – FA8

Complete Step-by-Step Solutions | Name: Francis Jhey Mago

Problem 1 (6 points)

The urn contains 10 red, 8 black, and 7 green balls, for a total of 25 balls. A color is chosen uniformly from the three colors, and then 4 balls are selected without replacement. Let X be the number of selected balls having the chosen color.

(a) Find P(X = 0).

By conditioning on the chosen color, the probability that none of the four selected balls has that color is:

$$P(X = 0) = (1/3)[C(15,4)/C(25,4) + C(17,4)/C(25,4) + C(18,4)/C(25,4)].$$

The numbers of balls not having the chosen color are 15 if red is chosen, 17 if black is chosen, and 18 if green is chosen. Also, C(25,4) = 12650, C(15,4) = 1365, C(17,4) = 2380, and C(18,4) = 3060.

$$P(X = 0) = (1/3)[(1365 + 2380 + 3060)/12650] = 6805/37950 \approx 0.1793.$$

Final answer: P(X = 0) = 0.1793.

(b) Find $P(X_i = 1)$, for i = 1, 2, 3, 4.

For any fixed position i, the probability that the ball in that position has the chosen color is the average of the three color proportions:

$$P(X_i = 1) = (1/3)(10/25 + 8/25 + 7/25) = (1/3)(25/25) = 1/3.$$

Final answer: $P(X_i = 1) = 1/3 \approx 0.3333$, for i = 1, 2, 3, 4.

(c) Find E[X].

Write X as the sum of the four indicator random variables: $X = X_1 + X_2 + X_3 + X_4$. By linearity of expectation:

$$E[X] = E[X_1] + E[X_2] + E[X_3] + E[X_4] = 4(1/3) = 4/3.$$

Final answer: E[X] = 4/3 ≈ 1.3333.

Problem 2 (8 points)

The Yule–Simon probability mass function is P(X = n) = 4/[n(n + 1)(n + 2)], for n ≥ 1.

(a) Show that this is a probability mass function.

For n ≥ 1, the numerator and denominator are positive, so P(X = n) ≥ 0. It remains to show that the probabilities sum to 1. Using the given identity:

$$1/[n(n + 1)(n + 2)] = (1/2)[1/\{n(n + 1)\} - 1/\{(n + 1)(n + 2)\}].$$

Therefore,

$$\Sigma_{n=1}^{\infty} P(X = n) = 2\Sigma_{n=1}^{\infty} [1/\{n(n + 1)\} - 1/\{(n + 1)(n + 2)\}].$$

$$= 2[(1/2 - 1/6) + (1/6 - 1/12) + (1/12 - 1/20) + \cdots].$$

The series telescopes: all intermediate terms cancel, leaving 2(1/2) = 1. Thus the function is a valid PMF.

Final answer: $\Sigma_{n=1}^{\infty} P(X = n) = 1$.

(b) Show that E[X] = 2.

By the definition of expected value:

$E[X] = \Sigma_{n=1}^{\infty} nP(X = n) = 4\Sigma_{n=1}^{\infty} 1/[(n + 1)(n + 2)]$.

Since $1/[(n + 1)(n + 2)] = 1/(n + 1) - 1/(n + 2)$,

$E[X] = 4[(1/2 - 1/3) + (1/3 - 1/4) + (1/4 - 1/5) + \cdots]$.

This telescopes to $4(1/2) = 2$.

Final answer: E[X] = 2.

(c) Show that $E[X^2] = \infty$.

$E[X^2] = \Sigma_{n=1}^{\infty} n^2P(X = n) = 4\Sigma_{n=1}^{\infty} n/[(n + 1)(n + 2)]$.

Partial fractions give $n/[(n + 1)(n + 2)] = -1/(n + 1) + 2/(n + 2)$. The summand is asymptotic to $1/n$ as n becomes large.

Because the harmonic series $\Sigma_{n=1}^{\infty} 1/n$ diverges, the series for $E[X^2]$ also diverges.

Final answer: $E[X^2] = \infty$.

Problem 3 (4 points)

The density is $f(x) = a + bx^2$ for $0 \le x \le 1$, and 0 otherwise. Given $E[X] = 3/5$, find a and b.

Since f is a probability density function, its integral over its support must equal 1:

$\int_0^1 (a + bx^2)\, dx = 1$.

$a + b/3 = 1$. (1)

Also, the expected value is:

$E[X] = \int_0^1 x(a + bx^2)\, dx = a/2 + b/4 = 3/5$. (2)

From (1), $a = 1 - b/3$. Substitute into (2):

$(1 - b/3)/2 + b/4 = 3/5$

$1/2 - b/6 + b/4 = 3/5 \Rightarrow 1/2 + b/12 = 3/5$

$b/12 = 1/10 \Rightarrow b = 6/5$.

Then $a = 1 - (6/5)/3 = 1 - 2/5 = 3/5$.

Final answer: a = 3/5 = 0.6000 and b = 6/5 = 1.2000.

Problem 4 (4 points)

Let X have expected value μ and variance σ^2. Define $Y = (X - \mu)/\sigma$. Find E[Y] and Var(Y).

Using linearity of expectation:

$E[Y] = E[(X - \mu)/\sigma] = [E[X] - \mu]/\sigma = (\mu - \mu)/\sigma = 0$.

Using the variance scaling property $Var(cX + d) = c^2Var(X)$:

$Var(Y) = Var[(X - \mu)/\sigma] = (1/\sigma^2)Var(X - \mu)$

$= (1/\sigma^2)Var(X) = \sigma^2/\sigma^2 = 1$.

Final answers: E[Y] = 0 and Var(Y) = 1.

Problem 5 (8 points)

Let X and Y be independent Poisson random variables with parameters $\lambda_1 = 2$ and $\lambda_2 = 3$, respectively. Let $W = X + Y$.

(a) Derive the MGF M_W(t).

The MGF of a Poisson random variable with parameter λ is $M(t) = \exp[\lambda(e^t - 1)]$. Therefore:

$M_X(t) = \exp[2(e^t - 1)]$ and $M_Y(t) = \exp[3(e^t - 1)]$.

Since X and Y are independent, the MGF of their sum is the product of their MGFs:

$M_W(t) = M_X(t)M_Y(t)$

$= \exp[2(e^t - 1)] \exp[3(e^t - 1)] = \exp[5(e^t - 1)]$.

Final answer: $M_W(t) = \exp[5(e^t - 1)]$.

(b) Identify the probability distribution and parameter(s) of W.

The MGF of a Poisson random variable with parameter λ is $\exp[\lambda(e^t - 1)]$. Comparing this with $M_W(t) = \exp[5(e^t - 1)]$, W has a Poisson distribution with parameter 5.

Final answer: W ~ Poisson(5), so its parameter is $\lambda = 5$.

(c) Find E[W] and E[W²] by differentiating M_W(t).

Given $M_W(t) = \exp[5(e^t - 1)]$, differentiate once:

$M'_W(t) = 5e^t \exp[5(e^t - 1)]$.

At t = 0:

$E[W] = M'_W(0) = 5e^0 \exp[5(e^0 - 1)] = 5$.

Differentiate again using the product rule:

$M''_W(t) = 5e^t \exp[5(e^t - 1)] + 25e^{2t} \exp[5(e^t - 1)]$.

At t = 0:

$E[W^2] = M''_W(0) = 5 + 25 = 30$.

Final answers: E[W] = 5.0000 and $E[W^2] = 30.0000$.